

SeismoFNO: Physics-Grounded Neural Operators for Seismic Structural Response Prediction

A full walkthrough of the project — motivation, methods, experiments, and open problems

Raghvendra Singh Gahlot · B.Tech (Civil Engineering, Structural Engineering) · September 2026

Why I started this

I got interested in this after running a handful of nonlinear time-history analyses in OpenSeesPy for a coursework assignment and noticing how long a single run took to converge, even for a small five-story shear frame. Regional risk assessment, city-scale loss estimation, and design-optimisation workflows all need this kind of analysis run thousands of times over. That gap between what rigorous simulation costs and what applications need felt like the right place to ask whether a learned surrogate could help. What I didn't expect going in was how badly the standard recipe — Fourier Neural Operators on a fixed grid — breaks the moment you try to apply it to buildings that don't all have the same number of storeys. That failure, and the two fixes that followed it, is most of what this project is about. I've tried to be as honest about what still doesn't work as about what does.

1. Foundations: getting the ground truth right first

Before any surrogate modelling, the first stretch of this project was building a trustworthy simulation pipeline and confirming the physics was sound end-to-end. That meant:

- Validating single-degree-of-freedom elastoplastic dynamics and Bouc-Wen nonlinear hysteretic loops against known OpenSeesPy benchmarks (EXP1).
- Tracking continuous hysteretic energy dissipation $E_h(t) = \int f_{int} du$ as a physically meaningful validation target, not just displacement error (EXP2).
- Deciding, early on, that random train/test splits weren't good enough — a model can appear to generalise while just interpolating within structural families it's already seen. I switched to splitting by structural archetype and by earthquake identity instead (EXP3).

This groundwork mattered more than I expected. Almost every surprising result later in the project traced back to something about the representation or the physics, not to training hyperparameters.

2. Problem formulation

A multi-storey shear building subjected to horizontal ground acceleration $a_g(t)$ is governed by:

$$M \ddot{u}(t) + C \dot{u}(t) + f_{int}(u(t), \dot{u}(t)) = -M \mathbf{1} a_g(t)$$

where $M, C \in \mathbb{R}^{(N \times N)}$ are the mass and Rayleigh damping matrices; f_{int} is the (possibly nonlinear) restoring-force operator; $u(t) \in \mathbb{R}^N$ is the relative floor-displacement vector; $\mathbf{1} = [1, \dots, 1]^T$. The surrogate goal is to approximate the operator mapping ground acceleration and structural parameters to the full response trajectory. This mapping should ideally be resolution-invariant — evaluating at arbitrary discrete time points shouldn't require retraining when the time discretisation changes — which is part of why the operator-learning framing made sense.

3. Buildings, ground motions, and the partitioning scheme

Buildings were idealised as lumped-mass shear frames: storey mass 30,000 kg, inter-storey stiffness 15–60 MN/m, storey height 3.5 m, 5% Rayleigh damping tuned to the first two modal frequencies, and a bilinear elastoplastic restoring-force law with 5% kinematic hardening. Six archetypes span fundamental periods from 0.35 s to 1.40 s:

Archetype	Storeys	T ₁ (s)	Role
3S_T035	3	0.35	Training / ID
3S_T060	3	0.60	Training / ID
3S_T090	3	0.90	Training / ID
5S_T055	5	0.55	Training / ID
5S_T085	5	0.85	Training / ID – upper training limit
5S_T120	5	1.20	Held-out OOD-B – unseen structure, the main test

Ground motions were drawn from the PEER NGA-West2 database. Partitioning ran along two independent axes – structural archetype and earthquake identity – giving four evaluation partitions with genuinely different physical meaning:

Partition	Structural group	Earthquake group	n	What it tests
In-distribution (ID)	known archetypes, $T_1 \leq 0.85$ s	training records	120	Ordinary accuracy
OOD-A	known archetypes, $T_1 \leq 0.85$ s	held-out records	300	Unseen ground motion
OOD-B	unseen structural period – the main test	training records	240	Unseen structural period
OOD-C	held-out archetype ($T_1=1.20$ s)	held-out records	60	Combined structural + seismic extrapolation

4. First attempt: a fixed-grid Fourier Neural Operator, and why it failed

The natural starting point was a standard 2D Fourier Neural Operator (Li et al., 2021), which has worked well on regular Cartesian domains such as fluid-flow fields. I built a fixed 5×2048 input tensor — five storeys (the tallest archetype) by 2,048 time steps — and zero-padded shorter buildings to fill the unused storeys. Five-storey buildings, which naturally filled the grid, reached a reasonable 19.29% median relative L2 error. Three-storey buildings, padded across two zeros, reached a median error of **99.6%** — essentially garbage.

The reason is mathematically straightforward once you look at it in the frequency domain. A discrete spatial Fourier transform implicitly assumes periodic boundary conditions across the lattice. Padding a three-storey building with two zero floors creates a sharp step between the last real floor and the padded region, and the spectral convolution reads that step not as "nothing there" but as a high-frequency ringing source that leads corrupted spectral energy back into the real floors below. This wasn't a training problem; longer training and more layers didn't help, because the representation itself imposed an invalid assumption: a building is a small, irregular graph of masses connected by columns, not a padded image.

Representation	3-storey median relative L2
Fixed-grid FNO2D (EXP4)	99.6%
Topology-native GNO (EXP5)	22.1%

Moving from a fixed grid to a native graph representation resolves the zero-padding failure.

5. A topology-native fix: graph neural operators

The fix was to stop forcing buildings onto a shared grid. Each building now owns its own graph $G = (V, E)$: one node per real storey ($|V| = N_{\text{stories}}$), edges between physically adjacent floors carrying the column stiffness. Each graph block computes:

$$m_v^{(\ell+1)}(t) = \phi \left(h_v^{(\ell)}(t), \sum_{u \in N(v)} \psi(h_u^{(\ell)}(t), e_{uv}) \right)$$

followed by a 1D temporal Fourier spectral layer applied node-wise to handle the dynamics:

$$h_v^{(\ell+1)}(t) = \phi \left(W h^{(\ell+1)}(t) + F^{-1} [R_{\ell}(k) \cdot F(h_v^{(\ell+1)})(k)](t) \right)$$

This structure eliminates zero-padding artifacts while coupling spatial inter-storey information exchange with temporal wave propagation on a shared latent representation.

6. What graph topology alone can't fix

That was satisfying, but it uncovered a second, subtler problem. Evaluating the same unconditioned graph model on the held-out flexible archetype ($T_1 = 1.20$ s, well outside the training range of $T_1 \leq 0.85$ s) gave a peak displacement error of around 35.2%, and full-trajectory relative L2 error above 115%. Looking at the FFT of the predicted displacement records showed the model was oscillating at approximately 1.12 Hz — the edge of what it was trained on — rather than tracking the true resonance at 0.83 Hz.

When two sinusoidal signals of identical amplitude but slightly different frequencies are compared pointwise, they drift into phase opposition. Once phase opposition is reached, the L2 dif-

ference is mathematically ~141%. Modal conditioning can address the amplitude part of this — but it cannot warp the temporal frequency basis functions, which are still fixed and frozen per the whole record.

7. Conditioning on what we already know: modal invariants via FiLM

Given a building's mass and stiffness matrices, mode shapes and natural periods are computable before any shaking happens, by solving the generalised eigenvalue problem:

$$K \phi_i = \omega_i^2 M \phi_i, T_i = 2\pi/\omega_i$$

Because this only uses the known, pre-earthquake $[M]$ and $[K]$, it leaks no information about the future ground motion or response trajectory. Using Feature-wise Linear Modulation (Perez et al., 2018) to let it rescale the network's internal representations per layer:

$$[\gamma, \beta] = \text{MLP}(c), h^{(l+1)} = (1 + \gamma) \odot h^{(l)} + \beta$$

I compared an unconditioned baseline against two conditioned variants — one using the fundamental period T_1 only, one using the first three periods and frequencies together — and a falsification control described below.

Model	ID Peak	OOD-A Peak	OOD-B Peak	OOD-C Peak
EXP5 (unconditioned)	12.70%	15.86%	35.21%	38.66%
EXP6-B (T_1 only)	2.09%	8.81%	13.47%	14.36%
EXP6-C (multi-modal)	8.87%	7.63%	13.06%	17.01%
EXP6-D (shuffled control)	2.65%	8.94%	24.33%	36.16%

Peak-displacement error across all four partitions. Conditioning helps everywhere, but the gap is largest exactly where it matters most — the unseen-structure partitions.

Checking this wasn't just extra capacity. Before trusting the improvement, I wanted to rule out the boring explanation: maybe the FiLM branch just gave the network more parameters to work with, and any auxiliary vector — physically meaningful or not — would have helped similarly. I re-ran evaluation with the conditioning vectors shuffled across buildings within a batch, so each sample received a *different* building's modal descriptor. Performance under shuffling degraded to 24.33% on OOD-B and 36.16% on OOD-C — worse than either conditioned variant, though still somewhat better than the unconditioned baseline. That pattern is consistent with the model relying on physical correspondence rather than scalar capacity, though I'd stop short of calling it proof of causality — it's evidence, not certainty, and I'd want a dimensionally matched but physically meaningless control vector rather than a shuffled real one to state the Section 5 result more rigorously.

8. Computational performance

Timings were measured on Apple Silicon (MPS) with process-wide device locks to avoid benchmark noise from concurrent Metal command buffers:

Model / setting	Latency	Throughput	Speedup vs. OpenSeesPy
OpenSeesPy reference, batch 1	54.68 ms	18.3 sim/s	1.0×
Fixed-grid FNO2D (EXP4, batch 1)	6.60 ms	151.5 sim/s	8.3×
Topology GNO (EXP5, batch 1)	16.25 ms	61.5 sim/s	3.4×
T1-conditioned GNO (EXP6-B, batch 1)	21.45 ms	46.6 sim/s	2.6×
T1-conditioned GNO (EXP6-B, batch 32)	30.19 ms	1,040 sim/s	57.8× (throughput)

The fixed-grid model is fastest in isolation, which isn't surprising — it's also the model that doesn't work. Among the models that actually generalise, single-sample speedup (2.5–3.4×) is useful for interactive design tools, not transformative. The batched throughput is practically interesting for regional screening of tens of thousands of structures.

9. Architecture summary

Component	Description
Pre-earthquake modal engine	Eigenvalue decomposition of $[K, M] \rightarrow$ natural periods $T_{1..3}$ and frequencies $\omega_{1..3}$. Zero response data used.
Graph construction	$G = (V, E)$: $ V = N_{\text{stories}}$ nodes (floor slabs), edges = interstory columns. No padding.
FiLM conditioning	MLP maps modal invariants c to (γ, β) per layer. Applied to both graph MP and temporal spectral channels.
Spatiotemporal blocks	4 blocks: graph message-passing (spatial) + 1D Fourier conv (temporal), both FiLM-modulated.
Output	Floor displacements $u_i(t)$, interstory drift ratios $IDR_i(t)$, peak engineering demand parameters.

10. Where I'd take this next

- **State-space models (S4, Mamba):** Replace static 1D Fourier temporal layers with causal state-space models to address phase drift. This is the most pressing technical gap.
- **Neural ODEs:** Couple the graph spatial operator with a neural ODE integrator to track instantaneous stiffness degradation during yielding.
- **3D asymmetric structures:** Extend from 2D planar to 3D buildings with torsional response and bidirectional ground motion.
- **Online period tracking:** Update the modal conditioning dynamically as the structural period elongates during severe yielding.
- **Tighter falsification ablation:** Use a dimensionally matched but physically meaningless control vector rather than a shuffled real one to state the modal conditioning result more rigorously.

11. Reproducibility

Data partitions were checked programmatically for zero overlap in structure or earthquake identity across the four partitions before any model saw them. EXP4, EXP5, and EXP6 variants are frozen and hash-verified against the evaluation logs referenced above, and the repository's unit test suite — covering the numerical solvers, split disjointness, and model I/O shapes — passes in full at the time of writing. The codebase, evaluation scripts, and an interactive demo comparing live model output against OpenSeesPy ground truth are available in the repository.

References

Li, Z., Kovachki, N., Liu, B., Bhattacharya, K., Stuart, A., & Anandkumar, A. (2021). Fourier Neural Operator for Parametric Partial Differential Equations. *International Conference on Learning Representations (ICLR)*.

Perez, E., Strub, F., de Vries, H., Dumoulin, V., & Courville, A. (2018). FiLM: Visual Reasoning with a General Conditioning Layer. *AAAI Conference on Artificial Intelligence*.

Pacific Earthquake Engineering Research Center (PEER). NGA-West2 Ground Motion Database.

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